Spring 2010 / EC571

Instructor:Ajay Joshi

Exam 1

Date: February 22, 2010

Name : _____

BU ID : _____

Question	Maximum points	Points scored
1	20	
2	20	
3	20	
4	20	
5	20	
Total	100	

1. The exam consists of five questions
2. The duration of the exam is 2 hours
3. The maximum grade is 100 points
4. Please write your name on all pages
5. The solutions should be written in the spaces provided on the test sheet
6. If you need to use extra sheets of paper, write your name and BU ID on it, and clearly indicate the question number
7. Material that can fit on two sheets of paper can be used for cheat sheets
8. Material sharing is not allowed
9. Calculators are allowed
10. Boston University Academic Code of Conduct applies

Question 1

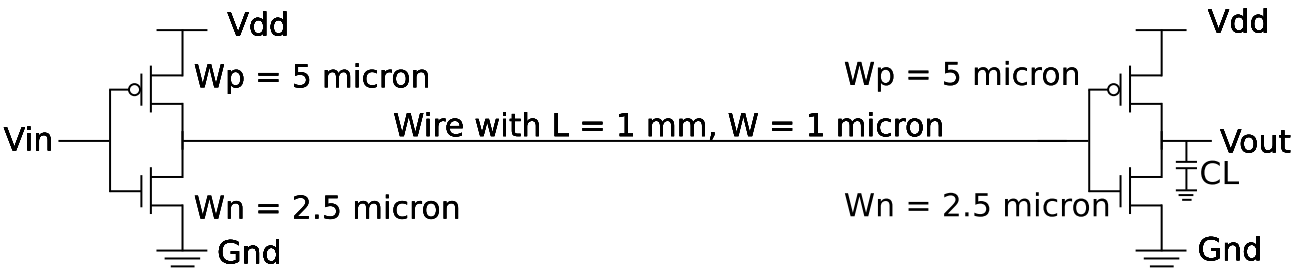


Figure 1

1. Draw the equivalent RC network. You can assume a lumped RC model for the wire

2. Write the Elmore delay equation for the design. Assume a step input at Vin

3. Using the following technology parameters calculate the total delay from Vin to Vout when Vin switches from 0 V to 2.5 V (V_{dd} = 2.5 V, L = 0.25 micron $R_{eq}(NMOS)$ = 13 Kohm/sq, $R_{eq}(PMOS)$ = 31 Kohm/sq, C_g = 1.5 fF/micron, C_d = 2 fF/micron, C_{wire} = 0.2 fF/micron, R_{wire} = 0.1 ohm/sq, C_L = 50 fF)

4. Calculate the average dynamic power dissipation of the circuit over 100 cycles. Assume 33 0 $\rightarrow$ 1 transitions and 33 1 $\rightarrow$ 0 transitions and an operating frequency of 1 GHz.

Question 2

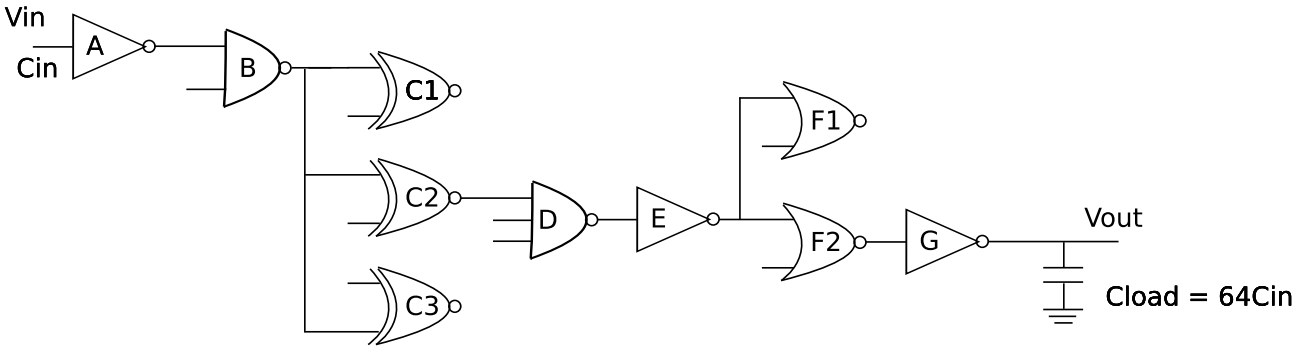


Figure 2

Using the logical effrot technique, determine the input capacitances for the various gates in the data path from Vin to Vout. You can assume that C1, C2 and C3 have the same sizes, and F1 and F2 also have the same sizes.

Question 3

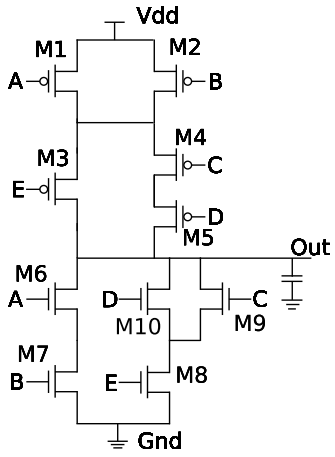


Figure 3

1. What logic is implemented by the circuit shown in the figure above?

2. Draw the equivalent RC network

3. Determine the size (width) of the various transistors relative to size of M1 so that worst case $\tau_{HL} = \tau_{LH}$. You can assume $2R_{eq}(NMOS) = R_{eq}(PMOS)$ and width of M1 transistor = W

4. Using the technology parameters in question 1.3, calculate worst case τ_{HL} and τ_{LH}. Assume width of M1 transistor W = 0.5 μm

Question 4

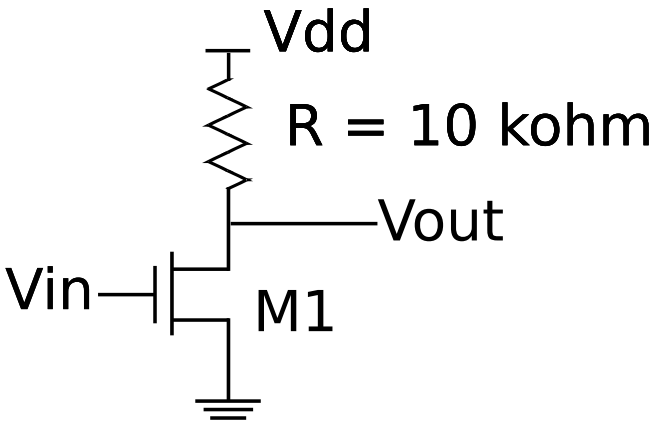


Figure 4

Assume V_{dd} = 2.5 V, $V_{TH,n}$ = 0.5 V and no short-channel effects.

1. For V_{in} = 0 V, what is the value of V_{out}?

2. For V_{in} = 2.5 V and V_{out} = 2.5 V, what is the region of operation for M1?

3. Write the equation for calculating V_{out} in that region

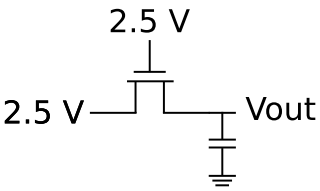
4. With V_{in} fixed at 2.5 V, V_{out} will start reducing. At what output voltage does the region of operation change for M1?

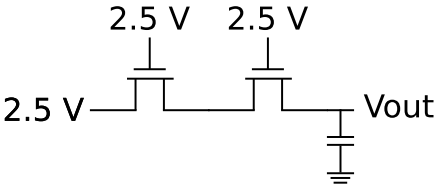
5. Write the equation for calculating V_{out} for this new region of operation

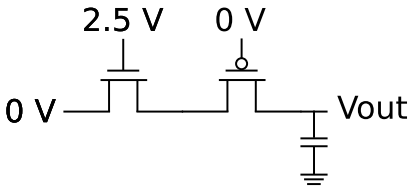
6. As V_{out} further decreases, what happens when $V_{out} << (V_{in} - V_{TH,n})$

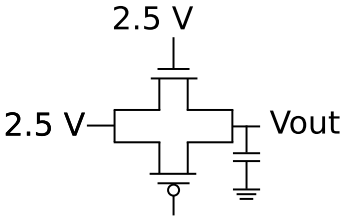
7. Write the equation for calculating V_{out} for this new region of operation

Question 5 Find the final value of the voltage V_{out} for the following 5 configurations. Assume V_{dd} = 2.5 V and abs(V_{th}) = 0.5 V. Assume that the capacitor is initially discharged. Ignore subthreshold conduction and body effect.









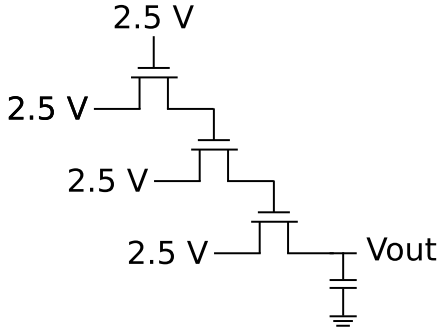


Figure 5